# Experiments on Tinystories

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## 1 Learning Rate Experiments

Tinystories dataset, fixed token budget of 32 * 256 * 500, evaluate at the last step=500 using 50 batches of size 32 each. Learning rates 1e-4, 3e-4, 1e-3, 3e-3, 1e-2, with cosine annealing schedule. Warm up iterations set at 5

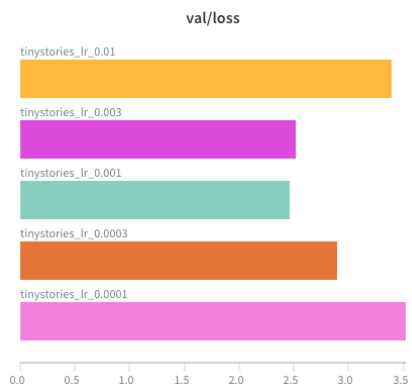



Figure 1:

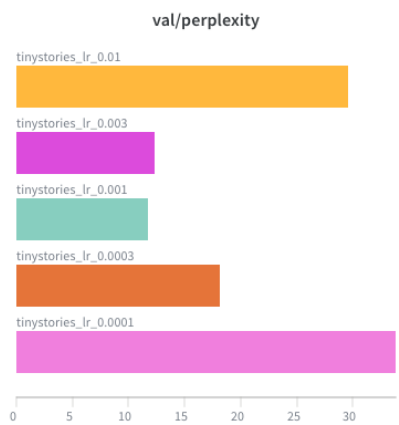
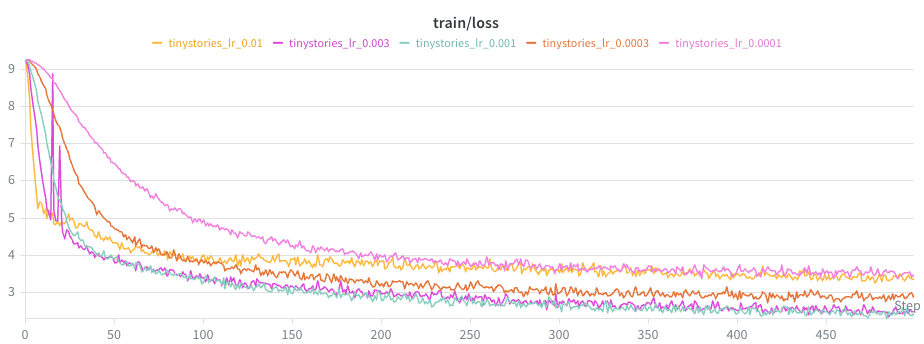
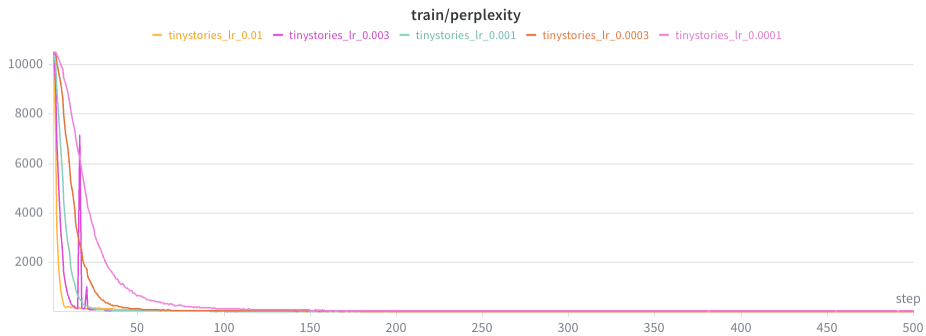
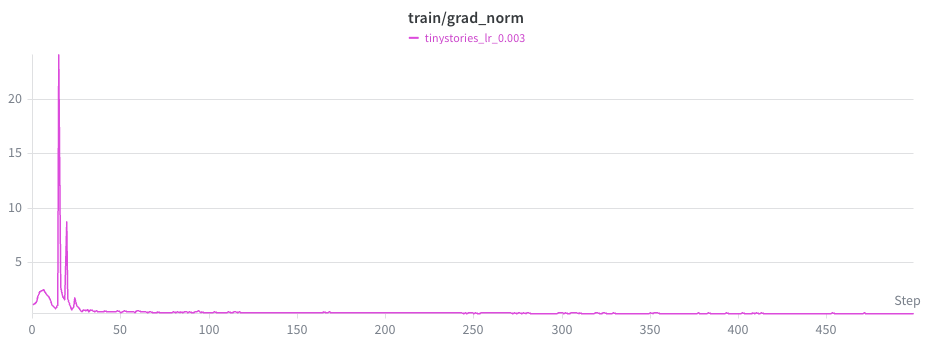


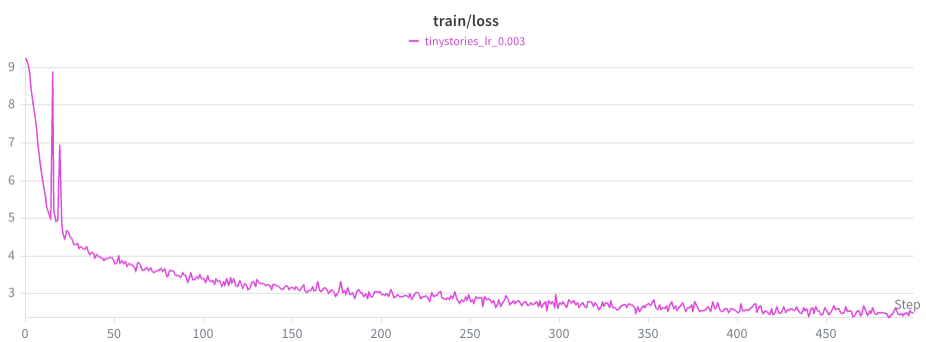

Figure 2:



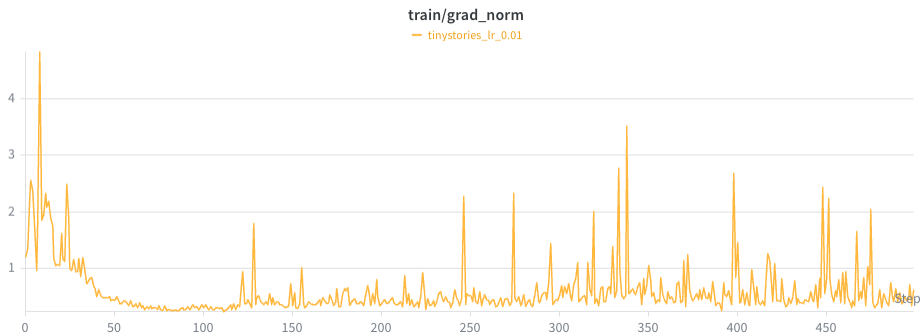


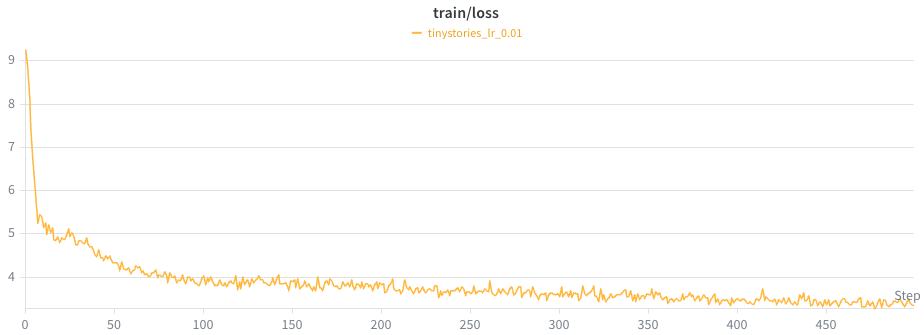

Gradient norm and training loss start to have spikes for lr=0.01 and 0.003 At LR=0.01, grad norm spikes are getting worse over time, which might eventually diverge with longer training. LR=0.003 trains faster initially but shows instability (spikes), LR=0.001 is more stable and achieves better final loss, LR=0.0003 and below are too conservative and underfit. The best LR sits just below where instability begins.

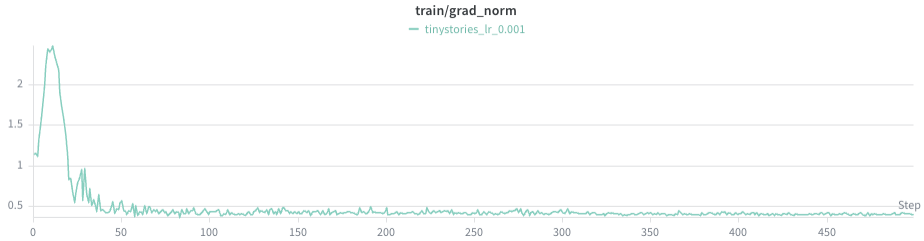



Best performance at LR=0.001



## 2 Batch Size Experiments

Fixed total tokens budget at 4096000; fixed validation tokens at 163840 Vary batch sizes: 4, 16, 32, 64, 128.

Best batch size: 4 (16 is smoother with similar validation loss)
Small batch sizes have noisier gradients and training losses, but there are more parameter updates and they can escape local minima. Larger batch sizes often are smoother but have fewer updates for the same amount of token budget. Larger batch often improves throughput because hardware is better utilized, but beyond some point, it can hurt.

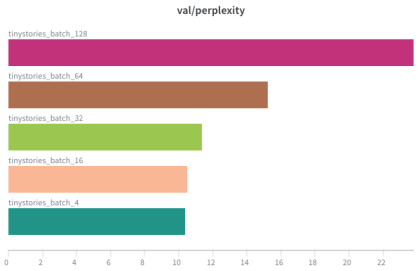



Figure 3:

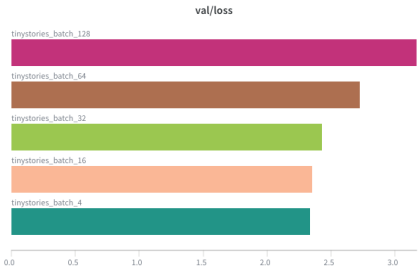
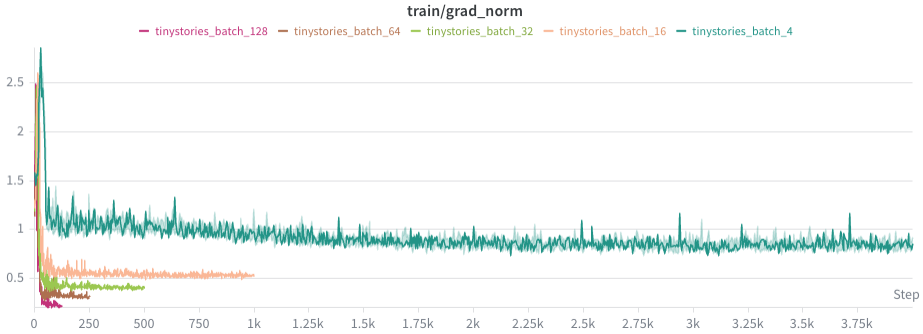
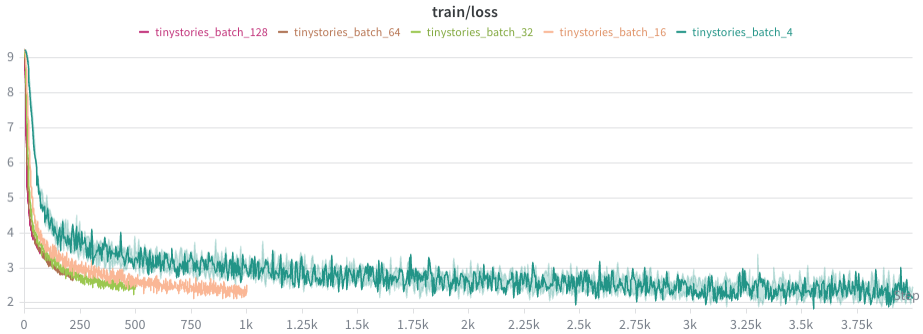
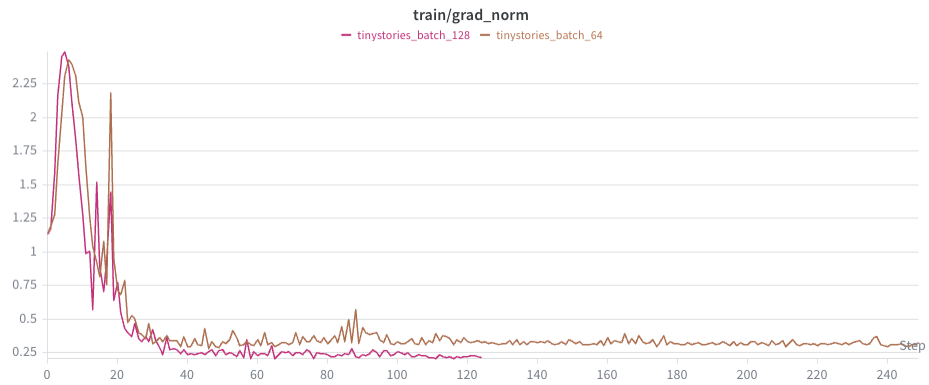


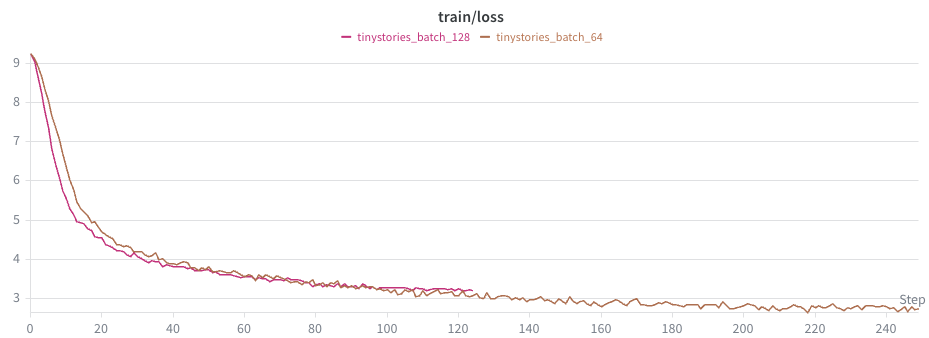

Figure 4:




We observe spikes in gradient norm for batch sizes 64 and 128 despite smoother training losses. We note that learning rate should scale linearly with batch size.



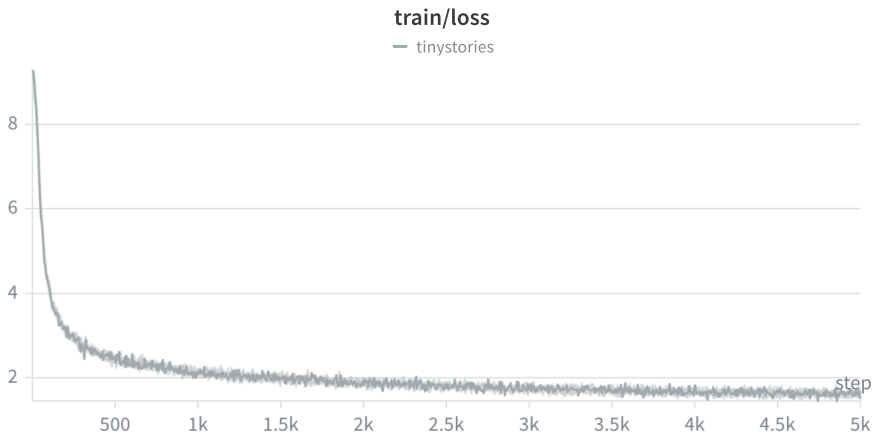



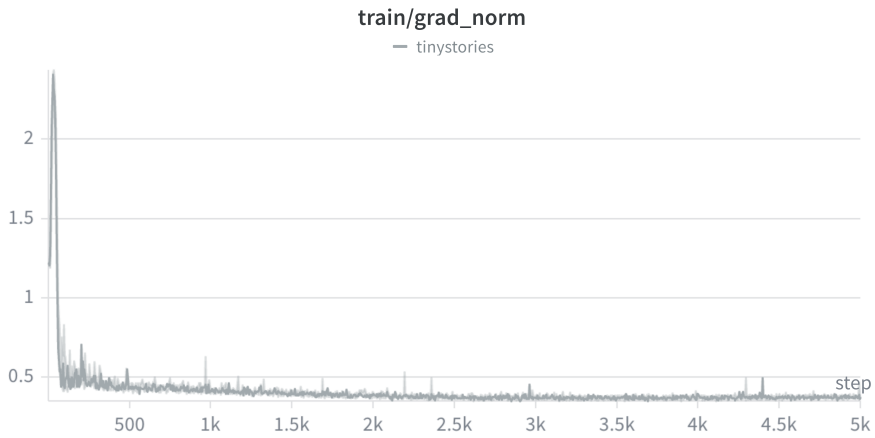

# 3 Full Run Results

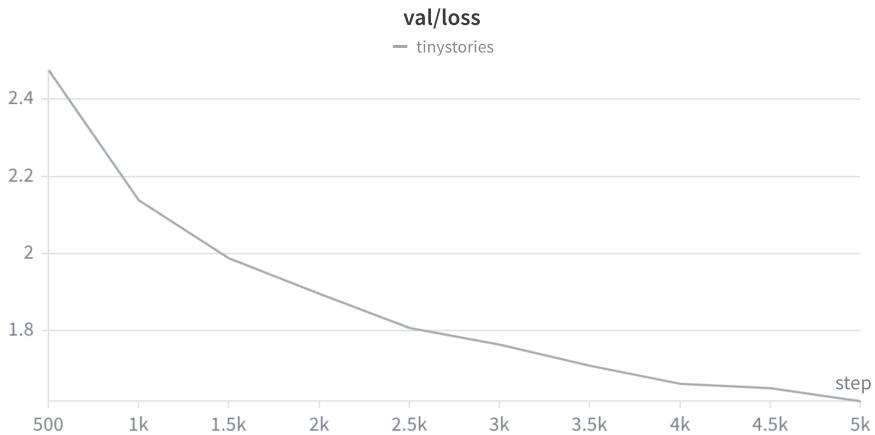
Parameters

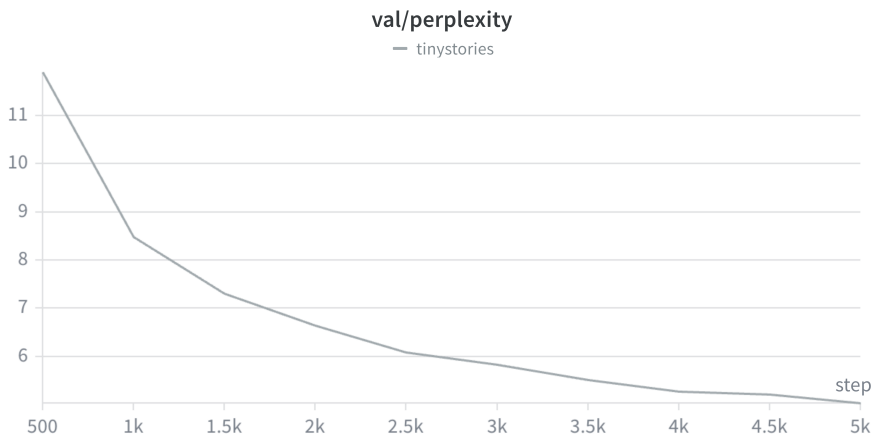
- Batch size: 32
- Context length: 256
- Max learning rate: 0.001
- Steps: 5000
- Validation batches: 50

Validation loss is 1.6 per token at 5000 steps.


train/loss
tinystories
8
6
4
2
step
500
1k
1.5k
2k
2.5k
3k
3.5k
4k
4.5k
5k


train/grad_norm
tinystories
2
1.5
1
0.5
step
500
1k
1.5k
2k
2.5k
3k
3.5k
4k
4.5k
5k


val/loss
tinystories
2.4
2.2
2
1.8
step
500
1k
1.5k
2k
2.5k
3k
3.5k
4k
4.5k
5k


val/perplexity
tinystories
11
10
9
8
7
6
step
500
1k
1.5k
2k
2.5k
3k
3.5k
4k
4.5k
5k